

MUSKEGON CGS, MI, USAWMO: **997341**Lat: **43.227N**Lon: **86.339W**Elev: **185**StdP: **99.12**Time Zone: **-5.00 (NAE)**Period: **05-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.4	-12.3	-18.7	0.7	-13.9	-16.6	0.9	-11.5	17.2	-5.6	15.7	-3.9	7.0	320	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.5	26.6	21.8	25.5	21.3	24.6	20.8	23.3	25.4	22.4	24.4	21.7	23.7	4.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.6	24.9	21.7	16.7	24.0	20.9	15.9	23.3	70.2	25.5	66.8	24.6	64.0	23.7	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	13.8	12.1	-17.4	28.7	3.2	1.3	-19.7	29.6	-21.6	30.3	-23.3	31.0	-25.6	31.9		
(5)			-17.5	24.1	3.5	1.2	-20.0	25.0	-22.1	25.7	-24.1	26.4	-26.6	27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-3.8	-4.2	0.0	7.0	12.4	17.7	20.5	20.4	17.0	10.7	5.1	0.3
	DBStd	9.59	4.77	4.29	4.38	3.86	4.04	3.16	3.04	2.54	3.23	3.81	4.17	3.99
	HDD10.0	1757	429	398	311	106	20	0	0	0	1	36	154	302
	HDD18.3	3748	687	632	569	341	188	49	14	11	62	238	397	560
	CDD10.0	1267	0	0	1	15	96	232	327	322	211	58	6	0
	CDD18.3	218	0	0	0	0	6	31	83	74	22	2	0	0
	CDH23.3	505	0	0	0	0	18	66	231	161	29	1	0	0
	CDH26.7	29	0	0	0	0	1	7	13	8	1	0	0	0
Wind	WSAvg	5.8	7.2	6.3	5.6	6.0	5.3	4.7	4.5	4.5	5.0	6.4	7.2	6.9
Precipitation	PrecAvg	889	57	45	59	87	89	88	77	80	80	90	80	59
	PrecMax	1102	129	92	125	190	239	166	140	146	163	209	185	156
	PrecMin	705	15	17	16	17	12	34	30	25	10	8	14	25
	PrecStd	121	26	22	32	35	47	34	30	32	34	48	43	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.8	6.0	14.4	19.8	25.9	27.7	28.0	27.7	25.7	22.8	15.8	11.8
		MCWB	5.4	5.0	7.9	12.2	18.7	19.6	23.3	21.9	20.1	19.0	11.9	11.0
	2%	DB	5.5	3.8	10.2	16.6	23.0	25.2	26.6	26.2	24.2	20.6	13.9	8.7
		MCWB	4.3	3.0	6.6	10.8	17.2	19.0	22.6	22.0	20.4	17.9	11.3	7.5
	5%	DB	4.2	2.5	7.5	14.5	20.5	23.6	25.6	25.1	23.0	18.7	12.4	7.1
		MCWB	3.1	1.4	4.8	10.1	15.3	19.1	22.1	21.4	20.0	16.2	10.1	5.9
	10%	DB	3.0	1.3	5.5	12.7	18.3	22.2	24.7	24.2	22.0	16.7	11.0	5.7
		MCWB	1.8	0.3	3.3	8.8	14.2	18.5	21.4	20.7	19.0	14.1	8.8	4.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	5.6	9.4	13.6	20.2	21.9	24.7	24.2	22.4	20.1	13.2	11.3
		MCDB	6.2	6.0	11.0	15.5	24.7	24.9	27.1	26.2	24.2	21.8	14.8	11.8
	2%	WB	4.7	3.2	7.2	11.9	17.9	20.7	23.7	23.1	21.6	18.2	11.5	7.6
		MCDB	5.2	3.7	9.4	14.7	22.0	23.3	25.9	25.2	23.4	19.5	13.5	8.4
	5%	WB	3.2	1.7	5.1	10.1	16.6	19.8	22.8	22.3	20.8	16.0	10.2	6.0
		MCDB	4.0	2.3	7.5	13.1	19.8	22.4	24.8	24.2	22.7	17.9	11.9	6.9
	10%	WB	1.8	0.3	3.5	8.8	15.1	19.0	21.9	21.5	19.6	14.4	9.0	4.7
		MCDB	2.9	1.1	5.3	11.7	17.7	21.4	24.0	23.4	21.2	16.3	10.7	5.6
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.5	6.1	6.7	6.8	6.1	5.1	5.5	6.2	5.7	5.2	4.3
		MCDBR	5.5	6.2	8.2	9.8	9.6	7.7	5.8	6.2	6.7	6.8	6.3	5.9
		MCWBR	5.5	6.3	5.8	5.8	5.8	4.6	3.6	3.9	4.2	5.7	4.8	5.7
	5% WB	MCDBR	5.2	6.4	7.3	8.3	8.7	6.1	4.9	4.4	4.8	6.7	5.9	5.7
		MCWBR	5.6	6.5	5.7	6.1	6.4	4.0	3.3	3.6	3.9	6.1	5.1	5.7
Clear-Sky Solar Irradiance	taub	0.298	0.325	0.341	0.372	0.403	0.417	0.418	0.414	0.390	0.353	0.319	0.297	
	taud	2.421	2.330	2.375	2.320	2.301	2.312	2.313	2.329	2.360	2.450	2.486	2.463	
	Ebn at Noon	837	866	899	893	871	856	851	843	835	827	807	806	
	Edh at Noon	76	98	106	122	128	127	126	120	108	87	71	67	
All-Sky Solar Radiation	RadAvg	1.18	2.01	3.57	4.77	5.61	6.27	6.38	5.43	4.31	2.65	1.52	0.96	
	RadStd	0.14	0.25	0.35	0.44	0.50	0.40	0.30	0.33	0.36	0.32	0.23	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page